Name: Shubham Patel
Date: 4/18/2024

Sign the statement below and attach it with your project. Your project will not be graded without it.

This project is entirely my work. I have not discussed this project with anybody in or out of class.

I did not use any unauthorized sources such as help from tutoring services (in person or online), nor have I used generative AI/Online assignment assisting websites (such as Chat GPT, DALL-E,Bard, Chegg, course Hero, etc.).

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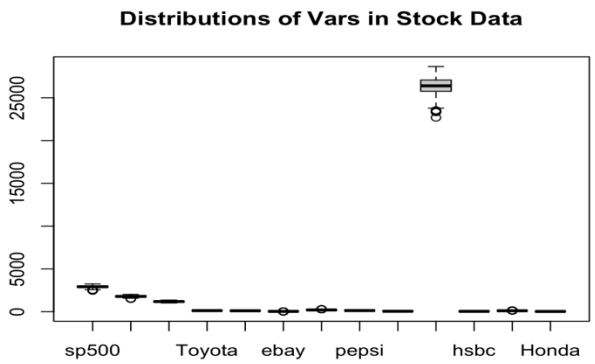
Project 3

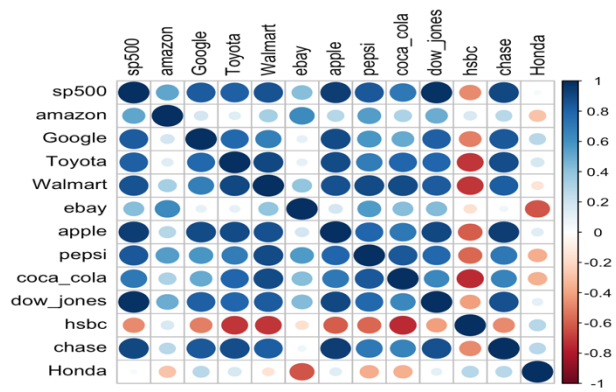
Name: Shubham Patel

Analysis A: Stock Data

In Analysis A, we are interested in figuring out if there are any common underlying latent variables that can explain/predict the behavior of the stock prices.



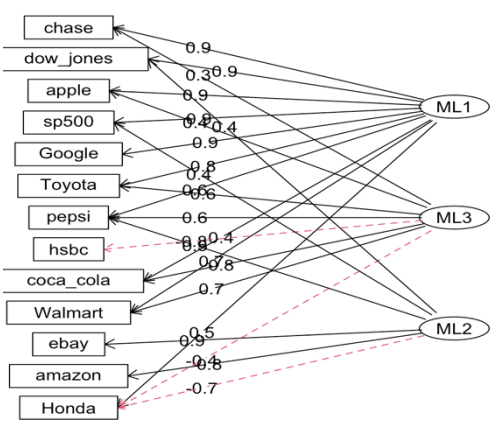
(Graph 1: Boxplot of stock data)



(Graph 2: correlation graph)

Based on the first graph, it seems that the variables in the dataset have differences in means and spreads. Before proceeding with factor analysis or principal component analysis, it's essential to standardize the variables to eliminate any potential bias introduced by differences in the scales of the variables. From the correlation graph, it seems that multiple variables are highly correlated, but we cannot predict clearly how many clusters of variables.

When we perform the likelihood ratio test, we receive a small p-value, but the **three-factor** model's residual matrix appears close to zero, so we took 3 as a number of factor and run MLE graph with varimax rotation.



(Graph 3: MLE varimax rotation)

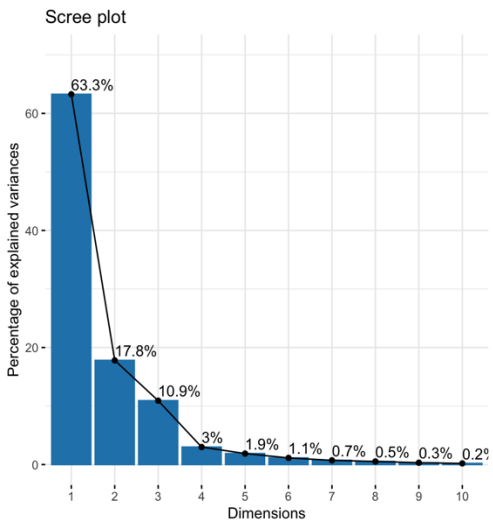
	ML1	ML2	ML3
sp500	0.994	−0.006	0.086
amazon	0.474	0.354	0.567
Google	0.840	−0.312	−0.080
Toyota	0.861	−0.103	−0.434
Walmart	0.902	0.222	−0.301
ebay	0.397	0.699	0.437
apple	0.958	−0.142	−0.166
pepsi	0.860	0.409	−0.064
coca_cola	0.751	0.431	−0.405
dow_jones	0.976	−0.060	0.148
hsbc	−0.532	−0.264	0.620
chase	0.921	−0.256	−0.176
Honda	0.037	−0.908	−0.016

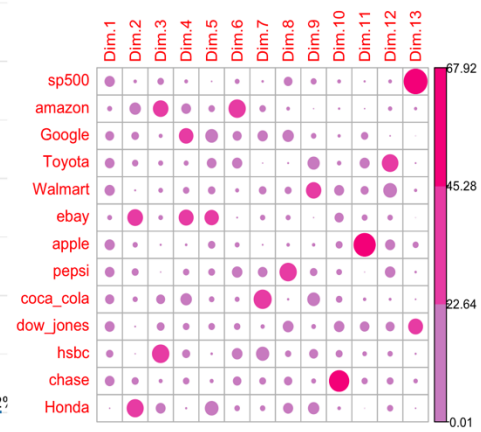
(Table 1: MLE loading)

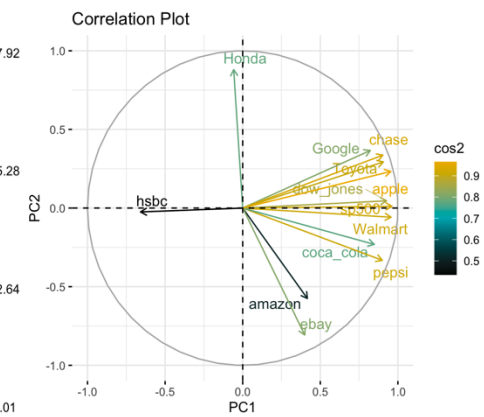
	V1	V2	V3
sp500	−0.962	−0.011	−0.241
amazon	−0.417	0.575	−0.625
Google	−0.822	−0.366	−0.130
Toyota	−0.907	−0.293	0.173
Walmart	−0.956	0.058	0.160
ebay	−0.400	0.807	−0.160
apple	−0.955	−0.233	−0.051
pepsi	−0.899	0.332	0.021
coca_cola	−0.848	0.231	0.345
dow_jones	−0.927	−0.046	−0.297
hsbc	0.659	0.025	−0.704
chase	−0.904	−0.337	−0.132
Honda	0.058	−0.880	−0.387

(Table 2: PC loading)

For MLE loadings, Factor 1 is the weighted average of all the stocks except HSBC. Factor 2 is the difference between the e-commerce business (Walmart, amazon and eBay) and beverage business (Pepsi and coca cola) with all other businesses (banks and car manufacturers). While on the other hand, the first factor from the PC loadings is weighted average of all the stocks except HSBC and Honda. Factor 2 is the same as the Factor 2 in the MLE loadings.





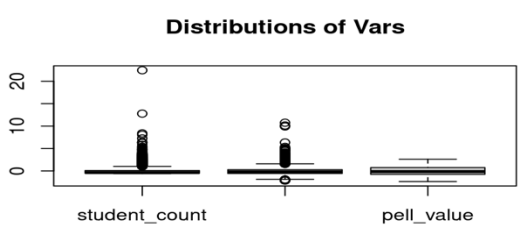


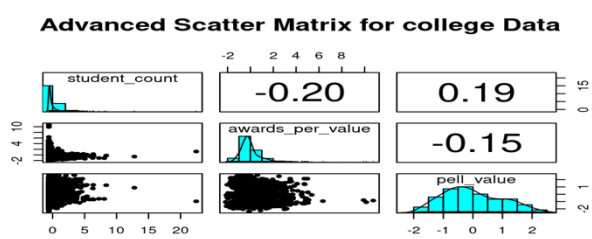
Based on the Scree plot, first **3 Principal components** account for ~92% of total variability in data. First Principal components summarize all the stock price except HSBC and Honda (Correlation plot), second principal components summarize eBay, Honda while based on correlation plot third principal components summarize Amazon, HSBC, coca cola and Dow jones.

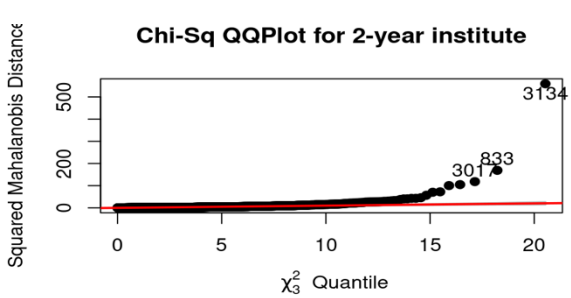
In comparing the results of factor analysis (FA) and principal component analysis (PCA) in Analysis A, Both FA and PCA agree on the number of latent variables or principal components required to explain the variability in the stock data. In this case, both methods indicate the need for three factors or principal components. There is alignment in the interpretation of the first two factors/components between FA and PCA. For example, Factor 1 in FA and PC1 in PCA summarize most stock prices, except HSBC and Honda.

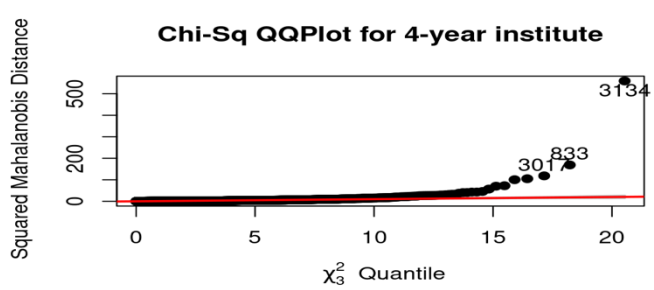
Analysis B1:

In Analysis B1, we are trying to determine if there are differences in number of students, award per value and Pell value between the 2-year degree institute and 4-year degree institute in US for academic year 2010-11. During this analysis, we are using a dataset which contains two categorical variables (level and controls) and three numerical variables (number of students, award per value and Pell value).







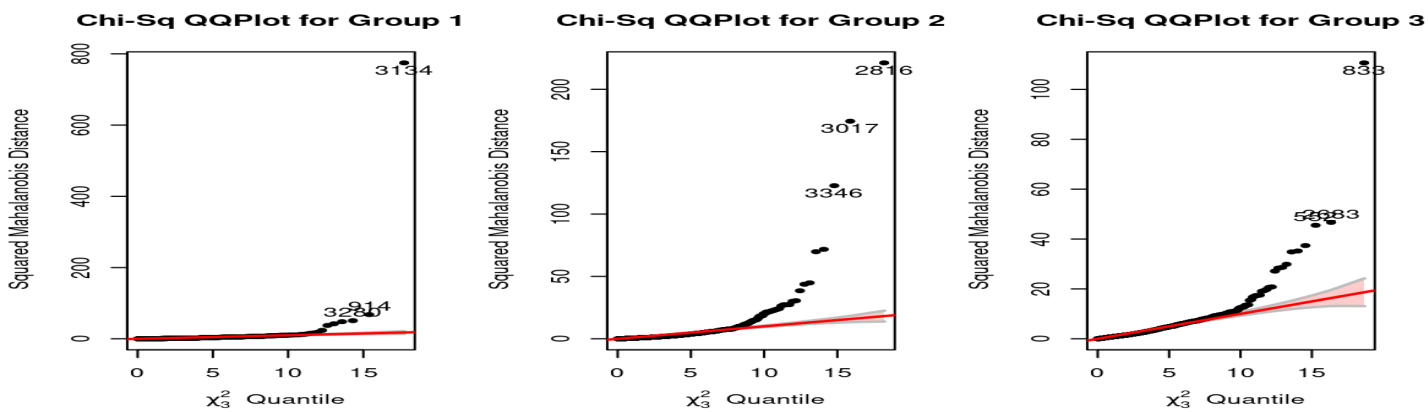


Firstly, we created a boxplot for all variables to see the range of all the variables for the given data. It suggests that we need to standardize our data. Secondly, we created an advanced scatter matrix to see the relationship between variables. Our scatter matrix suggests that Pell value and awards per value are normal, while student count showed skewness in data. Chi-sq QQ plots show that it's normal with some extreme outliers.

When we did statistical analysis, based on the given data, strongly suggest that there are no differences in the number of students, award per value, and Pell value between the 2-year degree institute and 4-year degree institute in the US for the academic year 2010-11. Because the P-value of the **two samples Hotelling's T^2** was large(P-value=1>0.05= $\boldsymbol{\alpha}$**)**

Analysis B2:

In Analysis B2, we are trying to determine if there are differences in number of students, award per value, and Pell value between the public, private not-for-profit, and private for-profit institutes in the US for the academic year 2010-11.



The chi-sq QQ plot indicates that given that is acceptable normal with some extreme outliers. When we did statistical analysis, we found based on the given data that there are very strong suggest that there is the difference in at least one variable (number of students, award per value, and Pell value) between the public, private not-for-profit, and private for-profit institutes in the US for the academic year 2010-11, Because the P-value of the **MANOVA Wilk's Lambda Test** was significantly small(P-Val=2.22e-16<0.05= $\boldsymbol{\alpha}$).

(Table 3: Multivariate Pairwise Comparisons)

	Wilks	Approx F	Num DF	Den DF	Pr(>F)
Private (Profit)– Private (Non-Profit)	0.8406	239.72	3	3792	2.2e-16
Private (Profit)– Public	0.6548	666.22	3	3792	2.2e-16
Private (Non-Profit) –Public	0.8378	244.38	3	3792	2.2e-16

Based on the above result, all three categories are different in at least one variable because all three Pairwise comparisons have P-values value less than 0.05 (Table 3).

(Table 4: Confidence Intervals using Bonferroni Correction)

	No of Student	Awards per value	Pell value
Private (Profit)– Private (Non-Profit)	(-1776.20, -224.78)	(6.503,8.764)	(-17.309, -12.898)
Private (Profit)– Public	(-8073.43, -6591.75)	(10.339,12.499)	(-19.679, -15.466)
Private (Non-Profit) –Public	(-7024.966, -5639.220)	(2.775,4.795)	(-4.439, -0.498)

Based on these 95% confidence intervals using Bonferroni correlation, there are significant differences in all three variables between the public, private not-for-profit, and private for-profit institutes in the US for the

academic year 2010-11, because all the possible values for the difference between all three groups for all three variables are non-zero at **95%** confidence interval method using **Bonferroni correlation (Table 4)**

R code :

Analysis A:

PCA:

```
#####################################################################
# Reading Data in R
#####################################################################
install.packages("readxl")
library(readxl)
# Read data
S2Data<- read_excel("StockData.xlsx")
head(StockData,5) # Displays firs 5 rows of data

# Extract only numerical variables for the analysis
SData = StockData[,2:14]
# Check for missing values. Rows with missing vals should be removed
colSums(is.na(SData))
# Keep only complete cases
SData = SData[complete.cases(SData),]
head(SData,5) # Displays firs 5 rows of data
# Visualize variables to check variances
dev.off()
boxplot(SData, main="Distributions of Vars in Stock Data")
# Compute numerical summaries for each varable
library(tableone)
SDataNumSum = CreateTableOne(data= SData,includeNA = FALSE)
print(SDataNumSum)
# Check original bi-variate scatterplots to see if obs (Stock) can be grouped
pairs(SData,main="Stock", pch=19)
# Standardize data
SData = as.data.frame(scale(SData))
```

```
SDataNumSum = CreateTableOne(data= SData,includeNA = FALSE)
print(SDataNumSum)

library(FactoMineR)
# Perform PCA analysis on standardized data
PCA.Res = PCA(SData, graph = FALSE,ncp = 13,scale.unit = TRUE)
# Getting the eigenvectors/loadings
sweep(PCA.Res$var$coord,2,sqrt(PCA.Res$eig[1:ncol(PCA.Res$var$coord),1]),FUN="/")
# See what information the PCA.Res object
PCA.Res
str(PCA.Res)
# Summary of PCA analysis
summary(PCA.Res)
# Alternative way of Computing loadings
PCA.Res1  = prcomp(SData,scale. = TRUE)
round(PCA.Res1$rotation,2)
# See what information the PCA.Res object
PCA.Res
str(PCA.Res)
# Summary of PCA analysis
summary(PCA.Res)
library("factoextra")
# Scree plot to visualize variance explained by each principal component
fviz_eig(PCA.Res, addlabels = TRUE, ylim = c(0, 70))
# Variable contribution to each PC
C1 = PCA.Res$var$contrib
round(C1,1)
# Quality of representation: Contribution of Vars to first 2 PCs
fviz_pca_var(PCA.Res, col.var = "cos2",
          gradient.cols =  c("black","#00AFBB", "#E7B800"),
          repel = TRUE, # Avoid text overlapping (slow if many points),
)+ labs(title ="Correlation Plot", x = "PC1", y = "PC2")
# Alternative way to see the contributions of each var to each PC
```

```
library("corrplot")
corrplot(PCA.Res$var$contrib, is.corr=FALSE, col= c("#C994C7","#DF65B0","#E7298A"))
```

Factor analysis:

```
#read the data
S2Data = read.xlsx(xlsxFile= 'StockData.xlsx')
head(S2Data,5) # Displays firs 5 rows of data
#keep only complete cases
S2Data=S2Data[complete.cases(S2Data),]
head(S2Data,5)#Display first 5 rows of data
# Extract only numerical variables for the analysis
S2Data = S2Data[,2:14]
#visualize variables to check variances
boxplot(S2Data,main="Distribution of vars in Stock Data")
#compute numerical summaries for each variable
library(tableone)
S2DataNumSum=CreateTableOne(data=S2Data,includeNA = FALSE)
print(S2DataNumSum)
#standardize data
S2Data=as.data.frame(scale(S2Data))
#computer correlation matrix
R=cor(S2Data)
#correlations between the vars in the study
library("corrplot")
corrplot(R,is.corr = TRUE,tl.col = "black")
#compute the loading and variable specific/unique Matrices
library(psych)
#Using Pcs
Eig.Res=eigen(R)
# L Matrix for m=3, the minus is so it matches the ones in the book
L = round(Eig.Res$vectors[,1:3]%*% diag(Eig.Res$values[1:3]^.5),3)
PSI = diag(diag(R - L%*%t(L)))
```

```
# Error Matrix
EPS = R - L%*%t(L) - PSI
# Communalities: row sum of squares
H = round(rowSums(L^2),3)
# Using the mle method
FA.Res = fa(S2Data, nfactors = 3, rotate = "none", fm = "ml")
L1 = round(FA.Res$loadings[,1:3],3)
PSI1 = diag(FA.Res$uniquenesses)
# Error Matrix
EPS1 = R - L1%*%t(L1) - PSI1
# Communalities: row sum of squares
H1 = FA.Res$communality
# Comparison of Rotations
# install.packages("GPArotation") # needed for the Quartimax Roation
library(GPArotation)
FA.Res.None = fa(S2Data, nfactors = 3, rotate = "none", fm = "ml")
FA.Res.Varimax = fa(S2Data, nfactors = 3, rotate = "varimax", fm = "ml")
FA.Res.Quartimax = fa(S2Data, nfactors = 3, rotate = "quartimax", fm = "ml")
par(mfrow = c(1, 3))
# The argument cut = 0.3 specifies that any loading less than 0.3 (in absolute value)
# will not be drawn, simple = F draws all the loadings (not just the largest
# loading per variable).
fa.diagram(FA.Res.None, cut = 0.3, simple = F, main = "No Rotation")
fa.diagram(FA.Res.Varimax, cut = 0.3, simple = F, main = "Varimax Rotation")
fa.diagram(FA.Res.Quartimax, cut = 0.3, simple = F, main = "Quartimax Rotation")

# Correlations Between the vars in the study
library("corrplot")
corrplot(R, is.corr=TRUE, col= c("#C994C7","#DF65B0","#E7298A"),tl.col="black")
# Test to find the appropriate number of factors:
pvals = c()
for(k in 1:5)
  pvals[k] = factanal(covmat = R,n.obs = 252,factors = k)$PVA
```

```
pvals
# Varimax rotation to the resulting model
FA.Res4 = fa(R, nfactors = 3, rotate = "varimax", fm = "ml")
L4 = FA.Res4$loadings[,1:3]
PSI4 = diag(FA.Res4$uniquenesses)
# Error Matrix
EPS4 = R - L4%*%t(L4) - PSI4
round(EPS4,3)
# Communalities: row sum of squares
H4 = FA.Res4$communality
# Visualize factors
fa.diagram(FA.Res4, cut = 0.32, simple = F, main = "Stock Data")
```

Analysis 2:

```
CData = read.xlsx(xlsxFile= 'PatelS-CollegeData(College_data).xlsx')
# Displays firs 5 rows of data
head(CData,5)
#covert first primary data into factor.
Cdata$level=factor(Cdata$level)
#standrized the data.
Cdata=cbind(Cdata[,1,drop=FALSE],as.data.frame(scale(Cdata[3:5])))
# Visualize variables to check thru box-plot
boxplot(X, main="Distributions of Vars")
X = Cdata[Cdata$level=="2-year",]
X = Cdata[,c(2,3,4)]
Y = Cdata[Cdata$level=="4-year",]
Y = Cdata[,c(2,3,4)]
#Scattermatrix
pairs.panels(Cdata[,2:4],main="Advanced Scatter Matrix for college Data",
        smooth = FALSE,
        ellipses = FALSE)
HotellingsT2Test(X,Y) # Hotelling’s test
```

```
HotellingsT2Test(X,Y, test = 'chi') # ChiSq test
cqplot(X, id.n=3,main = "Chi-Sq QQPlot for 2-year institute") # ChiSq QQPlot for 1st group
mvn(X, mvnTest = "hz")
cqplot(Y, id.n=3,main = "Chi-Sq QQPlot for 4-year institute") # ChiSq QQPlot for 2nd group
mvn(Y, mvnTest = "hz")
# Raw data
#MVN2Sample.HT.CIs.f(X,Y,conf.level= 0.95,var.eq = TR)
```

Analysis 3:

```
SData = read.xlsx(xlsxFile= 'PatelS-CollegeData(College_data).xlsx')
# Displays firs 5 rows of data
head(SData,5)
# Remove first columnns
SData = SData[, -1]
SData[,1] = as.factor(SData[,1])
# Computing Means and Var/Cov Mats per groups
VMeans = statList(SData[,-1],SData[,1],FUN = colMeans)
VMat = statList(SData[,-1],SData[,1],FUN = var)
Ns = table(SData[,1]) # Sample sizes
p = 3
g = 3
n = nrow(SData)
# Manually computing the W matrix
W = (Ns[1]-1)*VMat[[1]] + (Ns[2]-1)*VMat[[2]] + (Ns[3]-1)*VMat[[3]]
# Split data by group
SDataByGr = split(SData,SData[,1])
# Checking Multivariate normality for each group
par(mfrow = c(1, 3))
for(i in 1:g){
X = SDataByGr[[i]][,-1]
cqplot(X, id.n=3,main=paste0("Chi-Sq QQPlot for Group ", i))
print(mvn(X, mvnTest = "royston"))
```

```
print(mvn(X, mvnTest = "hz"))
}
# MANOVA with built in function
Y = as.matrix(SData[,-1])
Gr = as.factor(SData[,1])
# First fit a linear regression
LM.res = lm(Y~Gr)
# Call Manova, This command gives all Tests
# Wilk's lambda, Pillai, Hotelling-Lawley and Roy
SUM = summary(Manova(LM.res),"Wilks")
SUM
library(car)
# The W matrix using the build it functions
W = SUM$SSPE
# Checking which of the groups differ from each other
mvpaircomp(LM.res,factor1 = "Gr",test = "Wilks", adjust = "bonferroni")
# Checking each variable for differences across the groups
summary.aov(LM.res)
# Simultaneous CIs, Critical value for Bonferroni
k = p*g*(g-1)/2
ta = qt(.05/(2*k),df=n-g,lower.tail = FALSE)
# Simultaneous CIs
i = 2 # Group A
j = 3 # Group B
v = 2 # Variable
LL = VMeans[[i]] - VMeans[[j]] - ta * sqrt((1/Ns[i]+1/Ns[j])*diag(W)/(n-g))
UL = VMeans[[i]] - VMeans[[j]] + ta * sqrt((1/Ns[i]+1/Ns[j])*diag(W)/(n-g))
cbind(LL,UL)
```